

Agronomic Insights

The enhanced analysis reveals several critical patterns that significantly refine our understanding of rice yield determinants:

Growth Stage-Specific Radiation Dependency The radiation analysis shows a dramatic escalation in importance from Stage 1 ($r=0.11$, SHAP=0.015) to Stage 4 ($r=0.68$, SHAP=0.14). This 9-fold increase in predictive importance confirms that grain filling radiation is not just important—it's the dominant yield-determining factor. The strong correlation ($r=0.68$) indicates that higher radiation during grain filling directly translates to higher yields, reflecting the physiological reality that 70-80% of grain weight accumulates during this 35-day period through current photosynthesis.

Temperature Thresholds and Seasonal Patterns The temperature progression analysis reveals a bimodal pattern with peaks during reproductive stages (Max temp stages 2 and 4 at $\sim 29^{\circ}\text{C}$), which aligns with rice's phenological requirements. Early maximum temperatures (Max temp1, SHAP=0.075) are more influential than late-season temperatures, suggesting that achieving adequate heat units during tillering and early reproductive development sets yield potential, while grain filling is more radiation-limited than temperature-limited.

Soil Fertility Interaction Networks The correlation matrix reveals strong interconnections between soil variables: organic matter content correlates positively with carbon-nitrogen ratio ($r=0.66$) and available phosphorus ($r=0.79$), indicating these factors work synergistically rather than independently. The negative correlation between organic matter and some radiation variables ($r=-0.39$ to -0.57) suggests potential shading effects from excessive organic matter or associated vegetation management issues.

Weather Interaction During Critical Periods The precipitation-radiation interaction during grain filling shows an optimal zone where moderate precipitation (4-8mm) coincides with high radiation (18-22 units) to produce maximum yields. This reflects the need for adequate soil moisture to support photosynthesis without excessive cloudiness that reduces radiation. Fields with high precipitation but low radiation consistently show lower yields, confirming that radiation limitation overrides water abundance during grain filling.

Model Diagnostic Insights The residual analysis shows excellent model performance ($R^2=0.962$ training, 0.749 cross-validation) with normally distributed residuals,

indicating the identified relationships are robust and not artifacts of overfitting. The PCA analysis reveals that 47% of variance is captured in the first two components, suggesting the rice yield system has identifiable primary and secondary drivers rather than being governed by numerous equally important factors.

Variable Category Hierarchy Meteorological factors dominate with 54.8% of importance, followed by soil chemistry (22.1%), management (12.9%), and soil physical properties (10.1%). This hierarchy indicates that while farmers can influence soil and management factors, weather adaptation strategies should be the primary focus for yield optimization.

Specific Recommendations for Farmers

Precision Planting Date Optimization Calculate backwards 120 days from your region's peak solar radiation period (typically late summer/early autumn) to determine optimal transplanting dates. Install a simple solar radiation meter or use local weather station data to track daily radiation accumulation. Target grain filling to occur when daily radiation exceeds $18 \text{ MJ/m}^2/\text{day}$ for at least 25 of the 35-day period. In regions with variable weather, use 10-day weather forecasts to fine-tune transplanting within a 2-week optimal window.

Temperature-Based Growth Stage Management Monitor daily maximum temperatures during early growth and ensure they consistently exceed 25°C during the first 40 days post-transplanting. Use water depth manipulation to moderate soil temperatures—maintain 3-5cm depth during cool periods to capture solar heating, increase to 8-10cm during heat waves. If maximum temperatures drop below 22°C for more than 5 consecutive days during early growth, consider protective measures like clear plastic row covers or windbreaks.

Soil Organic Matter Optimization Strategy Target soil organic matter levels between 2.5-3.5% based on the correlation patterns observed. Above 3.5%, the negative correlations suggest diminishing returns and potential nitrogen immobilization issues. Incorporate rice straw annually but ensure C:N ratios below 25:1 by adding 25-30 kg N/ha during decomposition. Apply organic amendments 30-45 days before transplanting to allow proper decomposition and nutrient release synchronization.

Integrated Phosphorus-Organic Matter Management The strong correlation ($r=0.79$) between organic matter and phosphorus availability suggests managing these together rather than separately. Apply phosphorus fertilizers (20-30 kg $\text{P}_2\text{O}_5/\text{ha}$) along

with organic matter amendments to enhance P availability through organic acid chelation. Conduct annual soil tests and maintain available P above 15 mg/kg, applying additional P if organic matter levels are below 2.5%.

Weather-Responsive Water Management During grain filling, implement dynamic water management based on daily weather conditions. Maintain continuous flooding (5-8cm) when radiation is high and precipitation is low. During high-precipitation, low-radiation periods, allow fields to drain slightly (2-3cm depth) to prevent anaerobic stress while maintaining root zone moisture. Use simple rain gauges and visual cloud cover assessment to make daily water depth decisions.

Variety Selection for Local Climate Adaptation Choose varieties with 110-130 day maturity that position grain filling during your region's peak radiation season. Prioritize varieties with proven performance under your local temperature patterns—the analysis shows early-season temperature tolerance is more critical than late-season heat tolerance. Consider varieties with strong radiation use efficiency if you're in a region with variable sunshine during typical grain filling periods.

Monitoring and Adjustment Protocol Establish simple field monitoring using maximum-minimum thermometers, rain gauges, and weekly soil moisture checks. Record daily maximum temperatures during the first 40 days and radiation conditions (sunny/partly cloudy/cloudy) during grain filling. If cumulative radiation during grain filling appears insufficient, consider delayed harvest to allow maximum grain weight accumulation, even if it means accepting slightly higher moisture content at harvest.

Risk Mitigation for Weather Variability Develop contingency plans for suboptimal weather scenarios. If late-season radiation is poor, supplement with foliar potassium applications (0.5% K_2SO_4 solution) to improve grain filling efficiency under low-light conditions. For seasons with poor early temperature accumulation, consider split plantings 10-14 days apart to spread risk and potentially capture better weather windows for critical growth phases.

These recommendations work synergistically because optimal soil preparation enhances the plant's ability to respond to favorable weather conditions, while proper timing ensures that management inputs align with the most yield-responsive growth periods identified in the analysis.